



GOVERNMENT OF INDIA
O/o DIRECTOR GENERAL OF CIVIL AVIATION
Opp. Safdarjung Airport, New Delhi- 110003

DGCA-15032(03)/1/2023-DAS

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Air Safety Circular 02/2023

Subject: Flight Data Analysis Programme.

Technical, operational, human failures etc. induce hazards in aircraft operations. The aim of every flight safety programme is to address and control these hazards. One of the way to address these potential hazard is through flight data monitoring. Flight Data Analysis Programme (FDAP) is an effective tool for the safety assurance.

DGCA CAR Section-5 Series-F Part-II requires all scheduled operators and non-scheduled operators as part of their Safety Management System (SMS) to establish a Flight Data Analysis Program to determine hazards/ shortcomings in their operation of the aircraft.

The FDAP involves:

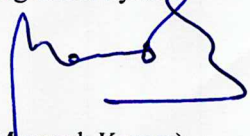
- Capturing and analyzing flight data to determine if the flight deviated from a standard operating envelope
- Identifying trends
- Communicating on findings and promoting action to reduce operational risks

In collaboration with the service providers and domain experts, pre-cursor events and their limit values have been identified for B787, B737, A320 family and ATR72 type of aircraft. This process would provide a standardized output for the management of key safety priorities i.e. Loss of Control, Runway Excursions and Controlled Flight into Terrain identified in National Aviation Safety Plan (NASP). In addition, it also provides a focused approach for the Safety Management System of an organization.

It is a continuous improvement process which captures experience of operators, regulator and any other agency. This process will be extended to cover other types of aircraft operating in India.

The pre-cursor events with their suggested limits for B787, B737, A320 family and ATR72 type of aircrafts are attached in Annexure 'A', 'B', 'C' and 'D' respectively.

These identified pre-cursor events are minimum, service providers should identify additional pre-cursor events catering to their type of operation for implementation of effective Safety Management System.


(Maneesh Kumar)
Joint Director General

Flight Exceedance parameters for B787

Sl. No.	Event Description	Low Limit	Medium Limit	High Limit	Reference
A. TAXI					
1	High speed Taxi Straight	31kts	33kts	35kts	SOP 2-9 (AI), Company limitation
2	High speed taxi in turn	10kts	12kts	15kts	Company limitation, FCTM 2.9 [For turn >90 deg. Further lower speed would require to open up more power]
B. TAKE-OFF					
3	Early Rotation	Vr-1kts	Vr-3kts	Vr-5kts	FCTM 3.10, 3.25
4	Late Rotation	Vr+10kts	Vr+12kts	Vr+15kts	FCTM 3.10, 3.25
5	Exceedance of Flaps/Slat Limit Speed	VFE -05kts	VFE -02kts	VFE -00kts	AFM Page 17 C. Limitation
6	Tyre Limit Speed High	194 Knots	199 Knots	204 Knots	FCOM B787-9 variant
7	Pitch High at Lift Off	7°	8°	9°	FCTM 3.8, IATA FDX value 9 B787-9 variant
		8°	9°	10°	FCTM 3.8, IATA FDX value 10 B787-8 variant
8	High Pitch Rate lift-off	3°/sec	3.5°/sec	4°/sec	FCTM 3.7

9	Low Pitch Rate at Take-off	1.75°/Sec	1.5°/Sec	1°/Sec	FCTM 3.7
10	Pitch Low at Lift Off	5°	4°	3°	FCTM 3.7 / 3.10
11	Excessive Bank Angle in Climb (Low ALT) T/O-100'	5°	5.5°	6°	FCTM 6.18 / FCTM 3.44
12	Max take-off weight	247.200 Tons	247.205 Tons	247.207 Tons	AFM / FCOM Limitation B787-9 variant
		227.930 Tons	227.930 Tons	227.935 Tons	AFM / FCOM Limitation B787-8 variant
13	Flap position during take-off			#5,10, 15,17,18,20	
14	Heading Deviation at Take Off from take-off roll 100 kts	3°	3.5°	4°	as per FDX
C. CLIMB					
15	Climb Out Speed Low 35'-1500'	V2+10kts	V2+5kts	V2+0kts	FCTM 3.44 [Acceleration height Vistara: 1500' Air India:1000']
16	VLO exceedance retraction	260kts/0.80 M	265kts/0.81 M	270kts / 0.82 M	AFM Page 15 C. Limitation
17	Pitch High in Climb GND TO FL 430	20°	23°	25°	Windshear/DGCA UPRT Circular: OC06/2018
18	Pitch Low in Initial Climb	10°	8°	6°	FCTM 3.7 / 3.10
19	Excessive Bank Angle in Climb (Med ALT) 100' to 1000'	20°	25°	30°	FCTM 6.18 /3.37, FCOM 11.1

20	Early Flap/Slat Retraction after Take Off	1500 ft	1300 ft	1000 ft	FCTM 3.45, Company policy for B787-9
		1000 ft	900 ft	800 ft	FCTM 3.45, Company policy for B787-8
21	Exceedance of Flap Altitude Limit	19800 ft	19900 ft	20000 ft	AFM / FCOM Limitation
D. CRUISE					
22	Speed Exceedance VMO	VMO -10 kts	VMO -5 kts	VMO -0kts	AFM Page 17 C. Limitation
23	Speed Exceedance MMO	MMO - .005	MMO -.002	MMO -.000	AFM Page 17 C. Limitation
24	Max Operating Alt Exceedance	43150 ft	43175 ft	43200 ft	AFM / FCOM Limitation
25	High acceleration in air			0/2 g	AFM Limitation
26	Level Bust	150 ft	250 ft	300 ft	RVSM Requirements
E. DESCENT					
27	High speed below FL100			FL100 - FL50 : 290 Kts Below FL50 : 250 kts	290 Kts based on turbulent penetration speed
F. APPROACH					
28	Max Landing Gear Extend Speed VLE	260kts	265kts	270kts / 0.82 M	AFM Page 17 C. Limitation
29	Gear Extension above VLO	260kts	265kts	270kts	AFM Page 17 C. Limitation
30	Approach Speed High (1000'-50')	Vref +10kts	Vref +12kts	Vref +15kts	OM(A) Pg-25-2 , Stabilized App

31	Approach Speed Low (1000'-50')	Vref + 0kts	Vref- 3kts	Vref - 5kts	Criteria, SOP 1-96 / OM(A) 25.9 FCTM 5.5
32	Flaps/Slats Limit SPD Exceedance	VFE -5kts	VFE -2kts	VFE -0kts	AFM Page 17 C. Limitation
33	Excessive Bank Angle 1000' - 0'	6°	7°	9°	FCTM 6.18
34	Excessive Bank Angle above 1000'	31°	33°	35°	FCOM Pg 11.1
35	High Rate of Descent (Med ALT) 1000'-30'	1000 ft/min	1050 ft/min	1100 ft/min	SOP 1-96, Stabilized App Criteria, OM(A) 25.9
36	Low ROD (Below 1000 feet)	<300 ft/min	<200 ft/min	<100 ft/min	
37	Late Landing Flap Setting	1200 ft	1150 ft	1100 ft	SOP 1-95-96, Stabilized App Criteria, OM(A) 25.9
38	Speedbrake on approach			Below 1000'	SOP 1-95-96, Stabilized App Criteria, OM(A) 25.9, FCOM limitation
39	Deviation below Glideslope 1000' - 200'	0.75 dot	0.85 dot	1.0 dot	SOP 1-96, Stabilized App Criteria, OM(A) 25.9
40	Deviation above Glideslope 1000' - 200'	-0.75 dot	-0.85 dot	-1.0 dot	SOP 1-96, Stabilized App Criteria, OM(A) 25.9
41	Deviation from Localizer 1000' - 50'	0.75 dot	0.85 dot	1.0 dot	SOP 1-96, Stabilized App Criteria, OM(A) 25.9

G. LANDING					
42	Speed High (Low ALT - 50' -20')	Vref + 8kts	Vref +10kts	Vref +15kts	
43	Speed Low (Low ALT - 50' -20')	Vref+0kts	Vref-5kts	Vref-10kts	
44	Significant Tail wind on Landing	9kts	10kts	15kts	FCOM I.10.2, FCOM I.10.4
45	Pitch High at Touchdown	5°	6°	7°	FCTM 6.11
46	Pitch Low at Touchdown	2°	1°	0.5°	FCTM 6.11
47	Pitch Rate High at Landing	2°/Sec	2.5°/Sec	3°/Sec	FCTM 6.9 / 6.11
48	Significant Roll In Flare Below 10 Feet	6°	7°	9°	SOP 1-96, Stabilized App Criteria, FCTM 6.18, 6.19
49	High Acceleration at Touch down	1.7 G	1.85 G	2 G	AFM Pg 14, C. Limitation
50	Land with Incorrect Flap Setting			#25,30	FCTM 5.8, FCOM 9.20.21
51	Heading Deviation at Landing above 60 Kts	3°	4°	5°	
52	Long Flare	8 sec	10 sec	12 sec	FCTM 6.9
53	Short Touchdown	300m	250m	200m	Aiming Point is 305m
54	Long Touchdown	700m	800m	925m	Touch Down Zone
55	Short Flare	5 sec	4 sec	3 sec	FCTM 6.9
56	Overweight Landing	192.700 Tons	192.750 Tons	192.776 Tons	AFM Limitation / Structure Limitation B787-9 variant

		172.365 Tons	172.370 Tons	172.370 Tons	AFM Limitation / Structure Limitation B787-8 variant
H. GENERAL / WARNINGS					
57	Excessive EGT at Take Off or Go Around	1035°C	1045°C	1065°C	AFM Pg 1, Engine C. Limitation
58	Max engine speed N1			108.50%	AFM Limitation
59	Max engine speed N2			117.50%	AFM Limitation
60	GPWS Caution/Warning		Caution	Warning	
61	Windshear warning			Warning	
62	TCAS RA			TCAS RA	
63	Stall warning			Warning	
64	Bounced landing			Bounce	
65	Go Around			Go around	Regulatory For Data Collection
66	Speed low during landing flap selection				Low Airspeed warning is covered in GPWS Airspeed Low warning

Flight Exceedance parameters for B737

SL.	Exceedance Parameters	Amber Band	Red Band
A.GENERAL			
1	Taxi Speed		>30 kts (Straight) & >40 kts (Runway Back Track)
2	Stick Shaker		When Triggered
3	Maximum N1		Operating Limit as per FCOM
4	Maximum N2		Operating Limit as per FCOM
5	Maximum EGT (Start)		Operating Limit as per FCOM
6	Maximum EGT (Takeoff)		Operating Limit as per FCOM
7	Maximum EGT (Continuous)		Operating Limit as per FCOM
8	Oil Temperature High		Operating Limit as per FCOM
9	Oil Pressure Low		Operating Limit as per FCOM
B. TAKEOFF			
1	Max Take-off Weight		As per FCOM
2	Pitch Rate High (Average)	>4.0 deg/sec	>4.5 deg/sec
3	Pitch Rate Low		<1.8 deg/sec
4	Speed of Rotation High	Airline may set their own value	>Vr+20 kts
5	Speed of Rotation Low	Airline may set their own value	<Vr-5 kts
6	Unstick Speed high	Airline may set their own value	>V2+25kts
7	Unstick Speed low	Airline may set their own value	<V2-10kts
8	Pitch Attitude High	Airline may set their own value	> FCOM- 1 Degree

9	Tyre Speed Limit	Airline may set their own value	As per FCOM
10	Early Autopilot Engagement	Airline may set their own value	As per FCOM (<400 ft)
11	Flap Position During Take-off	Airline may set their own value	As per FCOM
12	Configuration Change During Take-off	Airline may set their own value	Movement of flap during takeoff roll
13	RTO	Airline may set their own value	<80kts is low speed RTO >80kts is high speed RTO
C. CLIMB			
1	Excessive Bank (0-400 ft)	Airline may set their own value	>8 deg
2	Excessive Bank (>400 ft)	Airline may set their own value	> 33 deg >36 deg for aircraft with winglets
3	Early Configuration Change (Flaps)	Airline may set their own value	<800 ft
4	Initial Climb Height Loss (0-1000ft)	Airline may set their own value	EGPWS warning
5	Climb Out Speed Low (<35ft AGL)	Airline may set their own value	<V2
6	Climb Out Speed Low (35ft- 400ft AGL)	Airline may set their own value	<V2+5
7	Climb Out Speed Low (400ft-800ft AGL)	Airline may set their own value	<V2+15
8	Excessive time to 1000ft	Airline may set their own value	>50 sec
9	Gear Retraction Speed	Airline may set their own value	As per FCOM
D. CRUISE & DESCENT			
1	V _{Mo} Exceedance	>335Kts	>340Kts
2	M _{Mo} Exceedance	Airline may set their own value	As per FCOM
3	High Vertical Acceleration (Air)	Airline may set their own value	2G & 0G
4	Exceedance of Max. Operating Altitude	Airline may set their own value	As per FCOM (>41000ft)

5	TCAS Event	Airline may set their own value	Activation of RA
E. APPROACH & LANDING			
1	Excessive Bank (>500 ft)	>30 deg, for >3 sec	>35 deg, for >3 sec
2	Excessive Bank (500 ft to 100 ft)	>20 deg, for >3 sec	>25 deg, for >3 sec
3	Excessive Bank (<100 ft)	>8 deg, for >3 sec	>10 deg, for >3 sec
4	Flap placard Speed Exceedance (in Kts)	Flaps -800/-8 -900/-900ER -700	Flaps -800/-8 -900/-900ER -700
		1 245 245 245	1 250 250 250
		2 245 245 245	2 250 250 250
		5 245 245 245	5 250 250 250
		10 205 200 205	10 210 205 210
		15 195 195 190	15 200 200 195
		25 185 190 165	25 190 195 170
			30 175 180 165
			40 162/166 170 156
5	Flap/ Slat Altitude Exceedance	>19,800 ft	>20,000 ft
6	Late Landing Flap Selection	Airline may set their own value	<1000 ft
7	Gear Extended Speed	>315 kts or 0.815M	>320 kts or 0.82M
8	Gear Extension Speed	>265 kts or 0.815M	>270 kts or 0.82M
9	Approach Speed High (1000 ft to 500 ft)	>Vref+20 kts, for >3 sec	>Vref+25 kts, for >3 sec
10	Approach Speed High (500 ft to 50 ft)	>Vref+18 kts, for >3 sec	>Vref+20 kts, for >3 sec
11	Approach Speed High (<50 ft)	>Vref+15 kts, for >3 sec	>Vref+18 kts, for >3 sec
12	Approach speed high at touchdown	Airline may set their own value	>VRef+20kts
13	Approach Speed Low (<1000 ft)	<Vref-2 kts, for >3 sec	<Vref-5 kts, for >3 sec
14	Deviation Below Glideslope (<1000 ft)	Airline may set their own value	> 1 dot, for >3 sec
15	Deviation above Glideslope (<1000 ft)	Airline may set their own value	>= 1 dot

16	Localizer Deviation (<1000 ft)	Airline may set their own value	> 1 dot, for >3 sec
17	Pitch Attitude High at Touch Down (-700)	>13.2 deg	>13.7 deg
18	Pitch Attitude High at Touch Down (-800/-8 MAX)	>9.5 deg	>10 deg
19	Pitch Attitude High at Touch Down (-900/-900ER)	>8.5 deg	>9 deg
20	Pitch Attitude Low at Touch Down	<0.5 deg	<0.3 deg
21	Low on approach between 180 & 120 Sec to touch down	Airline may set their own value	<1000ft
22	Low on approach between 90 & 60 Sec to touch down	Airline may set their own value	<500ft
23	Speed Exceedence below 10000Ft	Airline may set their own value	>250kts Or State requirement
24	High Normal Acceleration at Touch Down	≥2 g	>2.1G (8samples) >2.2G (16 Samples) As per Boeing AMM
25	GPWS Caution/Warning	Caution	Warning
26	Speed Brake on Approach	Airline may set their own value	<1000 ft AGL
27	High ROD (1000-500 ft)	1200 Ft/min (Yellow) 1300 Ft/min (Amber)	1400 Ft/min
28	High ROD (500-100 ft)	1000 Ft/min (Yellow) 1050 Ft/min (Amber)	1100 Ft/min
29	Low ROD (<1000 ft)	Airline may set their own value	<300 fpm, for 3 sec
30	Late Thrust Reverser Deployment	Airline may set their own value	>10 Sec
31	Max. Landing Weight	Airline may set their own value	A/c Specific Weight
32	Long Landing	Airline may set their own value	>3000 feet or 30% of the LDA

Flight Exceedance parameters for A320

S.No.	Parameter	Time frame	Finalized Values		
			Yellow	Amber	Red
A. GENERAL					
1	Max engine speed: N1 N2		Operating limit as per FCOM Operating limit as per FCOM		
2	EGT Start on ground		Operating limit as per FCOM		
3	EGT in flight		Operating limit as per FCOM		
4	EGT Max. Continuous		Operating limit as per FCOM		
5	EPR 1 & 2		Operating limit as per FCOM		
6	Max. T/O Wt.		Operating limit as per FCOM		
7	Oil Temperature High		Operating limit as per FCOM		
8	Taxi Speed	-	32	35	40
B. TAKEOFF					
1	Pitch rate high on take off		4.5	5.5	6.5
2	Pitch rate low on take off		2.25	2	1.5
3	Unstick speed high				>V2+20 kts
4	Unstick speed Low				<V2-8 kts
5	Pitch attitude high during take off		10	11	12
6	Tyre speed limit high on take off				195 kts
C. CLIMB					
1	Excess Banking (0-100ft)	> 3 sec	6	8	10
2	Excess Banking (100-400ft)	> 3 sec	15	20	25
3	Excess Banking (400-1000ft)	> 5 sec	25	30	35
4	Early configuration change after T/O		< 900 ft	<800 ft	<700ft
5	Climb out speed low (100 ft- 1500ft AGL)		< V2+6	< V2+2	< V2-2
6	Gears retraction speed				>220 kts
D. CRUISE/DESCENT					
1	VMo exceedance			VMO + 4 kts >3Sec	VMO+4 Kts >6Sec

2	MMo exceedance			MMO+ 0.004 >3Sec	MMO+ 0.004 >6Sec
3	Max. OperatingAltitude				39800 ft
4	High vertical accn.		1.4G, 0.6G	1.6G, 0.4G	1.8G, 0.2G
E. APPROACH & LANDING					
1	Excess Banking (> 1000 ft)		32	36	40
2	Excess Banking(1000- 400ft)	> 5 sec	25	30	35
3	Excess Banking(400- 100ft)	> 3 sec	15	20	25
4	Excess Banking (100-0ft)	> 3 sec	6	8	10
5	Flap Placard Speed exceedance for different flap/slat position	> 3 sec		VFE	VFE+4Kts
6	Exceedance of flap/slat altitude				>20200 ft
7	Late landingflap (Flaps not in landing position)				IFR<1000 ft VFR<500 ft
8	Gear down speed				>250 kts
9	Approach speedhigh below 1000ft of Altitude		Vapp+15 kts	Vapp+20 kts	Vapp+25 kts
10	Approach speed high below 500 ft Altitude		Vapp+10 kts	Vapp+15 kts	Vapp+20 kts
11	Approach speed high below 50 ftAGL		Vapp+8 kts	Vapp+11 kts	Vapp+15 kts
12	Approach speed high at touchDown		Vapp+0	Vapp+5 kts	Vapp+10 kts
13	Deviation above/below glide slope (1000-200ft)	>3sec		±0.5	±1.0
14	Localiser Deviation	>3sec		± 0.5	±1.0
15	Approach speedlow below 1000 feet.		Vapp -5kts	Vapp -8kts	Vapp-10kts
16	Pitch Attitude High at Touch down.		8.2°	9.2°	10.2°
17	Pitch Attitude low at Touch down.		≤2°	≤1°	≤0.5°
18	Low on approach between <1000ft. to 200ft.		Glide Angle -0.4°	Glide Angle -0.6°	Glide Angle -0.8°
19	Speed exceedance below 10,000 ft.				≥ 250 Kts

20	High Normal acceleration on touch down.		≥1.7	≥1.85	≥2.0
21	GPWS/EGPWS warning triggered	Any time warning triggered			
22	Speed brake on approach.				<1000 ft AGL
23	High ROD(1000-500ft)		1200ft/min	1300 ft/min	1400 ft/min
24	High ROD(500-100ft)		1000 ft/min	1100 ft/min	1150 ft/min
25	Late reverser Deployment		5 Sec	10 Sec	20 Sec
26	Low ROD(Below 1000ft)				<400 ft/min
27	Long flare in time (Sec) 30 ft to touch down		10 sec	12 sec	14 sec
28	Long landing(Distance)		750 M	900 M	1050 M
29	Max. Landing Wt.	> Operating limit as per FCOM/ Performance limitation			

Flight Exceedance parameters for ATR72

SL NO	PARAMETER	FINAL LIMITS		Reference
		AMBER	RED	
PHASE	GENERAL			
1	Max engine speed NH1	-	106.4	FCOM, Revision 29.4 Lim 5, Section 70.1.5
2	Max engine speed NH2	-	106.4	FCOM, Revision 29.4 Lim 5, Section 70.1.5
3	Max ITT during Start on ground ENGINE 1	-	950 Deg	FCOM, Revision 29.4 Lim 5, Section 70.1.5
4	Max ITT during Start on ground ENGINE 2	-	950 Deg	FCOM, Revision 29.4 Lim 5, Section 70.1.5
5	Max ITT in flight ENGINE 1	-	800°C for 5 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
6	Max ITT in flight ENGINE 2	-	800°C for 5 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
7	ITT Max. Continuous ENGINE 1	-	800°C for 5 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
8	ITT Max. Continuous ENGINE 2	-	800°C for 5 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
9	Max. T/O Wt.	-	23000KG	FCOM, Revision 29.4 Lim 2, Section 1
10	Oil Temperature High ENGINE 1	-	140°C for 20 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
11	Oil Temperature High ENGINE 2	-	140°C for 20 sec	FCOM, Revision 29.4 Lim 5, Section 70.1.5
12	Flap Placard Speed exceedance flap -30	150 Kts	150 Kts 5 sec	FCOM, Revision 29.4 Lim 3, Section 1.3
13	Flap Placard Speed exceedance flap -15	185 Kts	185 Kts 5 sec	FCOM, Revision 29.4 Lim 3, Section 1.4

14	VLO Extension	-	170 Kts	
15	VLO Retraction	-	160 Kts	
16	VLE	-	185 Kts	
17	Taxi Speed High	35 Kts	40 Kts	
	Taxi Speed High on runway	-	50 Kts for 5 Sec	
PHASE	TAKE-OFF			
18	Pitch rate high on take-off	>4.5 Deg /Sec	>5 Deg / Sec	
19	Pitch rate low on take off	<1.5 Deg /Sec	<1.25 Deg / Sec	
20	Speed of Rotation High	V2+15 Kts	V2+20 Kts	
21	Speed of Rotation Low	-	V2-15 Kts	
22	Pitch attitude high during take-off(0 to 20 Ft)	-	9 Deg	
23	Tyre speed limit high on take-off	-	165kts	FCOM, Revision 29.4 Lim 3, Section 1.4
PHASE	CLIMB			
23	Excess Banking (20 to 400 ft)	8 Deg > 3 Sec	10 Deg > 3 Sec	
24	Excess Banking (> 400 ft. to 1000 Ft)	28 Deg > 5 Sec	30 Deg > 5 Sec	
25	Excess Banking (> 1000 ft.)	36 Deg > 5 Sec	40 Deg > 5 Sec	
26	Early configuration change after T/O	-	< Acceleration Altitude	FCOM
27	Initial climb height loss (20 to 1500 ft)	-	GPWS Don't sink Caution	FCOM, Revision 29.4, PRO.NOP, Section NSU 34.5
28	Climb out speed low (35FT TO Acceleration Altitude)	<=V2 Kts	<=V2 - 5 Kts	
29	Climb out speed low(Acceleration Altitude to 1500ft)	-	<=V2 +5 Kts	
PHASE	CRUISE / DESCENT			
30	VMO exceedance		>250kts	FCOM, Revision 29.4 Lim 3, Section 1.1
31	MMO exceedances		>0.55 M	FCOM, Revision 29.4 Lim 3, Section 1.1
32	High acceleration vertical	>1.8 G, <0.2 G	>2 G, <0 G	
33	High acceleration lateral	-	>ABS(0.35 G)	AMM
34	High ROD Top of Descent	-	3500 Ft / Min	

	to 2000 Ft			
35	Exceedances of max. operating altitude	-	25000 Ft	FCOM, Revision 29.4 Lim 1, Section 5
PHASE	APPROACH AND LANDING			
36	Excess Banking (1000 to 500 ft.)	30 Deg > 5 Sec	35 Deg > 5 Sec	
37	Excess Banking (500 ft to 100ft.)	15 Deg > 3 Sec	17 Deg > 3 Sec	
38	Excess Banking (100 ft to 0ft.)	10 Deg > 3 Sec	12 Deg > 3 Sec	
39	Flap Placard Speed exceedance flap -30	150 Kts	150 Kts 5 sec	FCOM, Revision 29.4 Lim 3, Section 1.3
40	Flap Placard Speed exceedance flap -15	185 Kts	185 Kts 5 sec	FCOM, Revision 29.4 Lim 3, Section 1.4
41	Late landing flap selection	-	Non-Precision Approach 1000 Ft AGL	
		-	ILS Approach 1000 Ft AGL	
		-	Visual Approach 500 Ft AGL	
42	Approach speed high below 1000 ft AGL	Vapp + 20 Kts for 3 Sec	Vapp + 25 Kts for 3 Sec	
43	Approach speed high below 500to50ft	Vapp + 15 Kts for 3 Sec	Vapp + 20 Kts for 3 Sec	
44	Approach speed high below 50 ft AGL	Vapp +11 Kts	Vapp +15 Kts	
45	Approach speed high at touch down	Vapp +5 kts	Vapp +10 kts	
46	Deviation above/below glide slope (1000 Ft - MDA)	>½ Dot>5 Sec	>1 Dot>5 Sec	
47	Localiser deviation (Alt. < 1000 feet)	>½ Dot>5 Sec	>1 Dot>5 Sec	
48	Approach speed low below 1000 feet.	VAPP - 2Kts for 5 Sec	VAPP - 5Kts for 3 Sec	
49	Pitch Attitude High at Touch down	6.5 Deg	7 Deg	
50	Pitch Attitude low at Touch down	-1.5 Deg	-2 Deg	
51	Pitch Rate High at Touch down	4°/sec	4.5°/sec	

52	Normal/Vertical acceleration High at touchdown	1.85 G	2.0 G	Manufacturer Recommendation
53	GPWS caution	-	Actual Caution triggered any time, Except Glideslope	FCOM, Revision 29.4, PRO.NOP, Section NSU 34.5
54	GPWS Operations - windshear		Not Applicable	FCOM, Revision 29.4, PRO.NOP, Section NSU 34.5
55	High ROD (1000-500 feet)	1200 Ft / Min for 5 Sec	1300 Ft / Min for 5 Sec	
56	High ROD (500-100) feet.	1000 Ft / Min for 3 Sec	1100 Ft / Min for 3 Sec	
57	Long bounce event	5Ft	6 Ft	
58	Low ROD (Below 1000 TO 50feet)	150 Ft / Min for 5 Sec	100 Ft / Min for 5 Sec	
59	Max. Landing Wt.	-	22350 kg	FCOM, Revision 29.4 Lim 2, Section 1
60	Long landing	-	3000 Ft (1/3rd of Runway Length for Airfield<9000 ft)	
61	Long Flare (TIME)	12 Sec	14 Sec	
62	Residual Power At Landing	15%	25%	FCOM PRO NOP.NOR.18